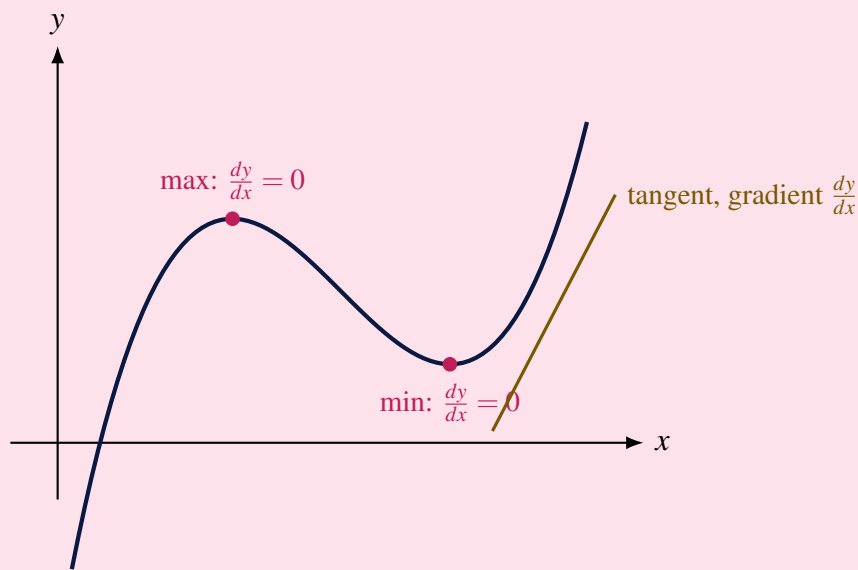


Differentiation 1

Cambridge IGCSE Mathematics 0580

Notes · Questions · Worked Answers

Power rule · Gradient at a point
Tangents & normals · Stationary points · Second derivative



How to use this booklet

- Read the pink **Key Notes** first. The **Index** lists every question, its source and its page.
- Sections 1–5: **Pemberton** Core & Extended book (Unit 10); the exercise and question number are shown above each question.
Sections 6–7: Cambridge **0580 exam questions 2020–2024, most recent first**, with the session, paper and question number.
-  No calculator = no calculator (Paper 2)  Calculator = calculator allowed (Paper 4).
- Each white **Question** page is followed by a yellow **Answer** page with working.

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Key Notes

1. The gradient function

$\frac{dy}{dx}$ (or $f'(x)$) gives the **gradient of the curve** (the gradient of the tangent) at any point.

Power rule: $y = ax^n \Rightarrow \frac{dy}{dx} = nax^{n-1}$ (multiply by the power, then reduce the power by 1)

• $\frac{d}{dx}(5) = 0$ (constant) $\frac{d}{dx}(7x) = 7$ $\frac{d}{dx}(2x^3 - 5x^2 + 6x) = 6x^2 - 10x + 6$

• Differentiate term by term. **Expand brackets** and **split fractions** first:

$$x(2x - 1)(x + 3) = 2x^3 + 5x^2 - 3x, \quad \frac{x^2 - 3}{x} = x - 3x^{-1}$$

• Write roots and fractions as powers: $\frac{1}{x^2} = x^{-2}$, $\sqrt{x} = x^{\frac{1}{2}}$, $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$, $\sqrt[3]{x^2} = x^{\frac{2}{3}}$

2. Using the gradient

- **Gradient at a point:** differentiate, then substitute the x -value.
- **Point(s) with a given gradient:** put $\frac{dy}{dx} = \text{value}$, solve for x , then find y from the curve.
- **Unknown constants a, b :** use each fact to make an equation (“passes through” uses y ; “gradient” uses $\frac{dy}{dx}$), then solve simultaneously.
- **Increasing** where $\frac{dy}{dx} > 0$, **decreasing** where $\frac{dy}{dx} < 0$ — solve the inequality (sketch the quadratic).
- Crosses the y -axis $\Rightarrow x = 0$; crosses the x -axis $\Rightarrow y = 0$.

3. Worked example

$$y = 2x^3 - 3x^2 - 12x + 5.$$

(a) $\frac{dy}{dx} = 6x^2 - 6x - 12$. Gradient at $x = 3$: $6(9) - 18 - 12 = 24$.

(b) Tangent at $(3, -4)$: $y + 4 = 24(x - 3) \Rightarrow y = 24x - 76$.

(c) Stationary points: $6(x - 2)(x + 1) = 0 \Rightarrow x = 2$ or $x = -1$; $(2, -15)$ and $(-1, 12)$.

(d) $\frac{d^2y}{dx^2} = 12x - 6$: at $x = 2$: $18 > 0$ **minimum**; at $x = -1$: $-18 < 0$ **maximum**.

4. Tangents and normals

At the point (x_1, y_1) , $m = \frac{dy}{dx}$ at x_1 .

- **Tangent:** $y - y_1 = m(x - x_1)$
- **Normal** (perpendicular to the tangent): gradient $-\frac{1}{m}$ (because $m_1 m_2 = -1$): $y - y_1 = -\frac{1}{m}(x - x_1)$
- If $m = 0$ the tangent is horizontal ($y = y_1$) and the normal is vertical ($x = x_1$).
- Angle with the x -axis: $\tan \theta = m$. Parallel lines have equal gradients.
- Where lines meet: solve simultaneously. Triangle areas: use the axis intercepts, $\frac{1}{2} \times \text{base} \times \text{height}$.

5. Stationary (turning) points

Stationary points occur where $\frac{dy}{dx} = 0$. Solve for x , then substitute into y to get the coordinates.

First derivative test (sign either side)

	left	point	right
maximum	+	0	−
minimum	−	0	+

Second derivative test

$$\frac{d^2y}{dx^2} < 0 \Rightarrow \text{maximum}$$

$$\frac{d^2y}{dx^2} > 0 \Rightarrow \text{minimum}$$

$$\frac{d^2y}{dx^2} = 0: \text{ use the first derivative test.}$$

$$\frac{d^2y}{dx^2} \text{ means differentiate } \frac{dy}{dx} \text{ again: } y = x^3 - 4x \Rightarrow \frac{dy}{dx} = 3x^2 - 4 \Rightarrow \frac{d^2y}{dx^2} = 6x.$$

No turning points: $\frac{dy}{dx} = 0$ has no real roots $\Rightarrow$ discriminant $b^2 - 4ac < 0$.

6. Exam traps

- $\frac{d}{dx}\left(\frac{1}{x^3}\right) = -3x^{-4}$, not $\frac{1}{3x^2}$. Never differentiate the top and bottom of a fraction separately.
- $(x + 5)^2 \neq x^2 + 25$ — expand the brackets before differentiating.
- Write the full coordinates (x, y) when asked for a point, not just x .
- Normal gradient is the **negative reciprocal**: $m = \frac{2}{3} \Rightarrow \text{normal } -\frac{3}{2}$.
- Give equations in the form asked (e.g. $y = mx + c$ or $ax + by + c = 0$ with integers).

1

No calculator

Pemberton Ex 10.4 · Q4–5

Use graphing software to investigate the value of the gradient at points on the curve $y = x^3$. Use this table to record your results.

x -coordinate of P	-4	-3	-2	-1	0	1	2	3	4
Gradient of curve at P									

Can you suggest a formula for the gradient of the curve $y = x^3$ at the point (x, x^3) ?

Use your formula from **question 4** to find the gradient of the curve $y = x^3$ at the points

a (5, 125)

b (-10, 1000)

c (0.2, 0.008)

2

No calculator

Pemberton Ex 10.5 · Q1

Differentiate with respect to x .

a x^6

b x^9

c x^{10}

d x^7

e 8

f x

g $\frac{1}{2}$

h -4

3

No calculator

Pemberton Ex 10.5 · Q2

Find $f'(x)$ for each of these functions.

a $f(x) = 3x^5$

b $f(x) = 2x^4$

c $f(x) = -5x^2$

d $f(x) = \frac{x^2}{2}$

e $f(x) = -\frac{2}{3}x^6$

f $f(x) = -5x$

g $f(x) = \frac{5}{4}x^4$

h $f(x) = -\frac{x^8}{4}$

1 Pemberton Ex 10.4 · Q4–5**Q4** For $y = x^3$:

x	−4	−3	−2	−1	0	1	2	3	4
grad	48	27	12	3	0	3	12	27	48

Formula: gradient = $3x^2$.**Q5** (a) $3(5)^2 = 75$ (b) $3(-10)^2 = 300$ (c) $3(0.2)^2 = 0.12$ **2** Pemberton Ex 10.5 · Q1

Multiply by the power, then subtract 1 from the power. A constant gives 0.

a $6x^5$ **c** $10x^9$ **e** 0 **g** 0**b** $9x^8$ **d** $7x^6$ **f** 1 **h** 0**3** Pemberton Ex 10.5 · Q2**a** $15x^4$ **c** $-10x$ **e** $-4x^5$ **g** $5x^3$ **b** $8x^3$ **d** x **f** -5 **h** $-2x^7$

4



No calculator

Pemberton Ex 10.5 · Q3

Find $\frac{dy}{dx}$ for each of the following.

a $y = 5x^2 - 2x + 4$

b $y = x^3 + 5x - 2$

c $y = 1 - 2x + 3x^2$

d $y = x(x + 2)$

e $y = 5x(3 - 2x^2)$

f $y = (x + 2)(x - 5)$

g $y = \frac{(x - 5)^2}{2}$

h $y = (2x^2 - 5)^2$

i $y = 5x(1 - x)^2$

4 *Pemberton Ex 10.5 · Q3*

Expand brackets first, then differentiate term by term. **a** $10x - 2$

b $3x^2 + 5$

c $-2 + 6x$

d $y = x^2 + 2x \Rightarrow 2x + 2$

e $y = 15x - 10x^3 \Rightarrow 15 - 30x^2$

f $y = x^2 - 3x - 10 \Rightarrow 2x - 3$

g $y = \frac{1}{2}(x^2 - 10x + 25) \Rightarrow x - 5$

h $y = 4x^4 - 20x^2 + 25 \Rightarrow 16x^3 - 40x$

i $y = 5x - 10x^2 + 5x^3 \Rightarrow 5 - 20x + 15x^2$

5



No calculator

Pemberton Ex 10.5 · Q4

Find the value of $\frac{dy}{dx}$ for each curve at the given point.

a $y = x^2 + 5x - 2$ at the point $(1, 4)$

b $y = 2 + 3x - 2x^4$ at the point $(0, 2)$

c $y = \frac{(x-2)^2}{4}$ at the point $-1, 2\frac{1}{4}$

d $y = (2x - 5)(x + 3)$ at the point $(2, -5)$

6



No calculator

Pemberton Ex 10.5 · Q5

Find the coordinates of the two points on the curve $y = x^3 + 2$ where the gradient is 12.

7



No calculator

Pemberton Ex 10.5 · Q6

Find the gradient of the curve $y = 3x^3 - 2x + 1$ at the point where the curve crosses the y -axis.

8



No calculator

Pemberton Ex 10.5 · Q7

Find the coordinates of the two points on the curve $y = x^3 - 5x^2 + 3$ where the gradient is 8.

5 Pemberton Ex 10.5 · Q4

a $2x + 5 \rightarrow 2(1) + 5 = \boxed{7}$

c $\frac{x-2}{2} \rightarrow \frac{-3}{2} = \boxed{-1.5}$

b $3 - 8x^3 \rightarrow \boxed{3}$

d $y = 2x^2 + x - 15; 4x + 1 \rightarrow \boxed{9}$

6 Pemberton Ex 10.5 · Q5

$$\frac{dy}{dx} = 3x^2 = 12 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2. \quad x = 2: y = 10; \quad x = -2: y = -6.$$

$$\boxed{(2, 10) \text{ and } (-2, -6)}$$

7 Pemberton Ex 10.5 · Q6

Crosses the y-axis at $x = 0$. $\frac{dy}{dx} = 9x^2 - 2 = \boxed{-2}$ at $x = 0$.

8 Pemberton Ex 10.5 · Q7

$$3x^2 - 10x = 8 \Rightarrow 3x^2 - 10x - 8 = 0 \Rightarrow (3x + 2)(x - 4) = 0$$

$$x = 4: y = 64 - 80 + 3 = -13; \quad x = -\frac{2}{3}: y = -\frac{8}{27} - \frac{20}{9} + 3 = \frac{13}{27}$$

$$\boxed{(4, -13) \text{ and } \left(-\frac{2}{3}, \frac{13}{27}\right)}$$

9



No calculator

Pemberton Ex 10.5 · Q8

Find the gradient of the curve $y = (2x - 3)^2$ at the point $(2, 1)$.

10



No calculator

Pemberton Ex 10.5 · Q9

- a Sketch the graph of $y = (x - 4)(x - 2)$ showing clearly any axis crossing points.
- b Find the gradient of the curve $y = (x - 4)(x - 2)$ at the points where the curve crosses the x -axis.

11



No calculator

Pemberton Ex 10.5 · Q10

- a Find the coordinates of the points where the curve $y = x^3 - 4x$ crosses the x -axis.
- b Find the gradient of the curve $y = x^3 - 4x$ at each of the points where the curve crosses the x -axis.

12



No calculator

Pemberton Ex 10.5 · Q11

The curve $y = x^2 - x + 1$ and the line $y = 2x + 5$ meet at the points A and B .

- a Find the coordinates of the points A and B .
- b Find the gradient of the curve at each of the points A and B .

9 Pemberton Ex 10.5 · Q8

$$y = 4x^2 - 12x + 9, \quad \frac{dy}{dx} = 8x - 12. \quad \text{At } x = 2: \text{ gradient} = \boxed{4}$$

10 Pemberton Ex 10.5 · Q9

(a) $y = x^2 - 6x + 8$: U-shaped, crosses x -axis at $\boxed{2}$ and $\boxed{4}$, y -axis at $\boxed{8}$, minimum at $(3, -1)$.

(b) $\frac{dy}{dx} = 2x - 6$: at $x = 2$: $\boxed{-2}$; at $x = 4$: $\boxed{2}$

11 Pemberton Ex 10.5 · Q10

(a) $x(x^2 - 4) = x(x - 2)(x + 2) = 0$: $\boxed{(-2, 0), (0, 0), (2, 0)}$

(b) $\frac{dy}{dx} = 3x^2 - 4$: $\boxed{8, -4, 8}$

12 Pemberton Ex 10.5 · Q11

(a) $x^2 - x + 1 = 2x + 5 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$:

$\boxed{A(-1, 3), B(4, 13)}$

(b) $\frac{dy}{dx} = 2x - 1$: at A : $\boxed{-3}$, at B : $\boxed{7}$

13



No calculator

Pemberton Ex 10.5 · Q12

The gradient of the curve $y = ax^2 + bx$ at the point $(3, 3)$ is 4. Find the value of a and the value of b .

14



No calculator

Pemberton Ex 10.5 · Q13

The gradient of the curve $y = ax^3 + bx^2 - 3$ at the point $(1, -5)$ is -2 . Find the value of a and the value of b .

15



No calculator

Pemberton Ex 10.5 · Q14

The curve $y = ax^3 + bx$ has gradient -1 when $x = 1$ and gradient -7 when $x = 0$. Find the value of a and the value of b .

16



No calculator

Pemberton Ex 10.5 · Q15

Given that the gradient of the curve $y = x^3 + ax^2 + bx$ is zero when $x = 1$ and when $x = 4$, find the value of a and the value of b .

17



No calculator

Pemberton Ex 10.5 · Q16

Given that $y = 3x^2 - 12x + 1$, find the range of values of x for which $\frac{dy}{dx} > 0$.

13 Pemberton Ex 10.5 · Q12

Point: $9a + 3b = 3 \Rightarrow 3a + b = 1$. Gradient $2ax + b$: $6a + b = 4$.

Subtract: $3a = 3$. $\boxed{a = 1, b = -2}$

14 Pemberton Ex 10.5 · Q13

Point: $a + b - 3 = -5 \Rightarrow a + b = -2$. Gradient $3ax^2 + 2bx$: $3a + 2b = -2$.

Solve: $\boxed{a = 2, b = -4}$

15 Pemberton Ex 10.5 · Q14

$\frac{dy}{dx} = 3ax^2 + b$. $x = 0$: $b = -7$. $x = 1$: $3a - 7 = -1 \Rightarrow a = 2$.

$\boxed{a = 2, b = -7}$

16 Pemberton Ex 10.5 · Q15

$\frac{dy}{dx} = 3x^2 + 2ax + b$. $3 + 2a + b = 0$ and $48 + 8a + b = 0$

Subtract: $45 + 6a = 0 \Rightarrow \boxed{a = -7.5, b = 12}$

17 Pemberton Ex 10.5 · Q16

$\frac{dy}{dx} = 6x - 12 > 0 \Rightarrow \boxed{x > 2}$

18



No calculator

Pemberton Ex 10.5 · Q17

A curve has equation $y = 3x^3 + 12x^2 + 16x$.

Show that the gradient of the curve is never negative.

18 *Pemberton Ex 10.5 · Q17*

$$\frac{dy}{dx} = 9x^2 + 24x + 16 = (3x + 4)^2.$$

A square is never negative, so the gradient is never negative.

19

No calculator

Pemberton Ex 10.6 · Q1

Find the equation of the tangent to each curve at the given point.

a $y = x^2 - 2x - 1$ at the point $(2, -1)$

b $y = x^3 - 5x - 3$ at the point $(1, -7)$

c $y = 5 - 3x^2 - x^4$ at the point $(-1, 1)$

d $y = x(5 - x)^2$ at the point $(2, 18)$

20

No calculator

Pemberton Ex 10.6 · Q2

A curve passes through the point $A(1, 1)$ and has equation $y = 3x^4 - 3x^2 + 1$.

Find the equation of the tangent to the curve at the point A .

21

Calculator

Pemberton Ex 10.6 · Q3

A curve has equation $y = 4x - 2x^2 - 2x^3$ and passes through the point $A(-1, -4)$.

a Find the equation of the tangent to the curve at A .

b Find the angle that this tangent makes with the x -axis.

22

Calculator

Pemberton Ex 10.6 · Q4

The curve $y = x^3 + 8$ intersects the x -axis at C . The tangent to the curve at C intersects the y -axis at D . Find the distance CD .

19 Pemberton Ex 10.6 · Q1

Tangent: $y - y_1 = m(x - x_1)$ with $m = \frac{dy}{dx}$ at the point.

(a) $m = 2x - 2 = 2$: $\boxed{y = 2x - 5}$

(b) $m = 3x^2 - 5 = -2$: $\boxed{y = -2x - 5}$

(c) $m = -6x - 4x^3 = 10$: $\boxed{y = 10x + 11}$

(d) $y = 25x - 10x^2 + x^3$, $m = 25 - 20x + 3x^2 = -3$: $\boxed{y = -3x + 24}$

20 Pemberton Ex 10.6 · Q2

$$\frac{dy}{dx} = 12x^3 - 6x = 6 \text{ at } x = 1. \quad y - 1 = 6(x - 1) \Rightarrow \boxed{y = 6x - 5}$$

21 Pemberton Ex 10.6 · Q3

(a) $\frac{dy}{dx} = 4 - 4x - 6x^2 = 2$ at $x = -1$. $y + 4 = 2(x + 1) \Rightarrow \boxed{y = 2x - 2}$

(b) $\tan \theta = 2 \Rightarrow \theta = \boxed{63.4^\circ}$

22 Pemberton Ex 10.6 · Q4

$x^3 + 8 = 0 \Rightarrow C(-2, 0)$. Gradient $3x^2 = 12$. Tangent $y = 12x + 24$, so $D(0, 24)$.

$$CD = \sqrt{2^2 + 24^2} = \sqrt{580} = \boxed{24.1}$$

23

No calculator

Pemberton Ex 10.6 · Q5

The equation of a curve is $y = 2x^3 - x^2 + 4$.

- a Show that the equation of the tangent to the curve at the point $(1, 5)$ is $y = 4x + 1$.
- b Find the coordinates of the point at which the tangent meets the curve again.

24

No calculator

Pemberton Ex 10.6 · Q6

The tangent to the curve $y = x^4 - 7x + 9$ at the point $(1, 3)$ intersects the x -axis at the point A and the y -axis at the point B .

- a Find the coordinates of A and B .
- b Find the area of triangle OAB , where O is the origin.

25

No calculator

Pemberton Ex 10.6 · Q7

The tangent to the curve $y = x^3 - 4x + 4$ at the point $(-2, 4)$ meets the x -axis at A and the y -axis at B . Find the midpoint of AB .

23 Pemberton Ex 10.6 · Q5

(a) $\frac{dy}{dx} = 6x^2 - 2x = 4$ at $x = 1$: $y - 5 = 4(x - 1) \Rightarrow y = 4x + 1$ ✓

(b) $2x^3 - x^2 + 4 = 4x + 1 \Rightarrow 2x^3 - x^2 - 4x + 3 = 0 \Rightarrow (x - 1)^2(2x + 3) = 0$

$x = -1.5, y = -5$: $\boxed{(-1\frac{1}{2}, -5)}$

24 Pemberton Ex 10.6 · Q6

$\frac{dy}{dx} = 4x^3 - 7 = -3$. Tangent: $y - 3 = -3(x - 1) \Rightarrow y = -3x + 6$

(a) $\boxed{A(2, 0), B(0, 6)}$ (b) Area = $\frac{1}{2} \times 2 \times 6 = \boxed{6 \text{ units}^2}$

25 Pemberton Ex 10.6 · Q7

$\frac{dy}{dx} = 3x^2 - 4 = 8$. Tangent: $y - 4 = 8(x + 2) \Rightarrow y = 8x + 20$.

$A(-2.5, 0), B(0, 20)$. Midpoint $\boxed{(-1\frac{1}{4}, 10)}$

26



No calculator

Pemberton Ex 10.6 · Q8

A curve has equation $y = 2x^3 - 4x^2 + 5$ and passes through the point $P(2, 5)$.

- a Find the equation of the tangent to this curve at the point P .
- b Find the coordinates of the point where the tangent at P meets the curve again. The tangent at the point Q is parallel to the tangent at P .
- c Find the coordinates of Q .

27



No calculator

Pemberton Ex 10.6 · Q9

The tangents to the curve $y = 2 - 3x - x^2$ at the points $(1, -2)$ and $(-3, 2)$ meet at the point P . Find the coordinates of P .

28



No calculator

Pemberton Ex 10.6 · Q10

The tangents to the curve $y = 2x + 3x^2 - x^4$ at the points $(-2, -8)$ and $(2, 0)$ meet at the point R . Find the coordinates of R .

29



No calculator

Pemberton Ex 10.6 · Q11

The curve $y = x(x - 1)(x - 5)$ crosses the x -axis at the points $O(0, 0)$, $A(1, 0)$ and $B(5, 0)$. The tangents to the curve at the points A and B meet at the point C . Find the coordinates of the point C .

26 Pemberton Ex 10.6 · Q8

$$(a) \frac{dy}{dx} = 6x^2 - 8x = 8: \boxed{y = 8x - 11}$$

$$(b) 2x^3 - 4x^2 + 5 = 8x - 11 \Rightarrow x^3 - 2x^2 - 4x + 8 = 0 \Rightarrow (x-2)^2(x+2) = 0: \boxed{(-2, -27)}$$

$$(c) 6x^2 - 8x = 8 \Rightarrow (3x+2)(x-2) = 0, x = -\frac{2}{3}: \boxed{Q\left(-\frac{2}{3}, 2\frac{17}{27}\right)}$$

27 Pemberton Ex 10.6 · Q9

$$\frac{dy}{dx} = -3 - 2x. \text{ At } (1, -2): m = -5, y = -5x + 3. \text{ At } (-3, 2): m = 3, y = 3x + 11.$$

$$-5x + 3 = 3x + 11 \Rightarrow x = -1: \boxed{P(-1, 8)}$$

28 Pemberton Ex 10.6 · Q10

$$\frac{dy}{dx} = 2 + 6x - 4x^3. \text{ At } (-2, -8): m = 22, y = 22x + 36. \text{ At } (2, 0): m = -18, y = -18x + 36.$$

$$22x + 36 = -18x + 36 \Rightarrow x = 0: \boxed{R(0, 36)}$$

29 Pemberton Ex 10.6 · Q11

$$y = x^3 - 6x^2 + 5x, \frac{dy}{dx} = 3x^2 - 12x + 5.$$

$$\text{At } A(1, 0): m = -4, y = -4x + 4. \text{ At } B(5, 0): m = 20, y = 20x - 100.$$

$$-4x + 4 = 20x - 100 \Rightarrow x = \frac{13}{3}: \boxed{C\left(4\frac{1}{3}, -13\frac{1}{3}\right)}$$

30



No calculator

Pemberton Ex 10.7 · Q1

The curve with equation $y = x^2 - 6x$ has one turning point

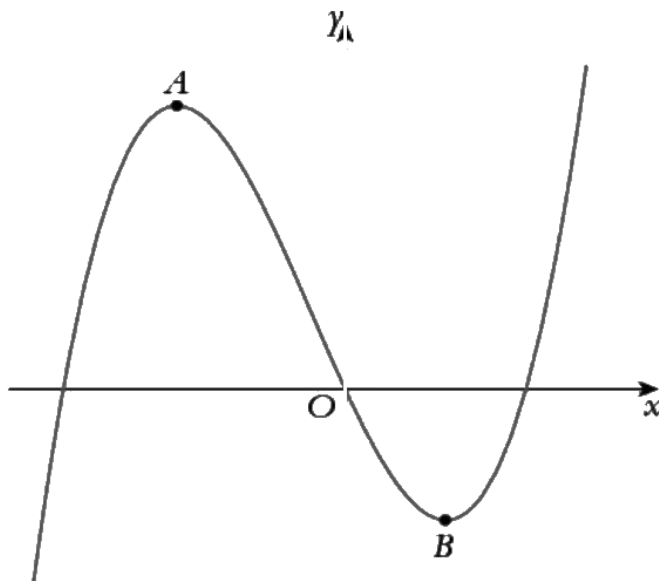
- Use completing the square to find the coordinates of the turning point and sketch the graph of $y = x^2 - 6x$. State whether the turning point is a maximum or a minimum point.
- Use differentiation to find the coordinates of the turning point.

31



No calculator

Pemberton Ex 10.7 · Q2



The diagram shows the graph of $y = x^3 + 3x^2 - 45x$.

The curve has a maximum point at A and a minimum point at B .

Find the coordinates of the points A and B .

30 Pemberton Ex 10.7 · Q1

(a) $y = (x - 3)^2 - 9$: turning point $(3, -9)$, a **minimum** (U-shape through O and $(6, 0)$).

(b) $\frac{dy}{dx} = 2x - 6 = 0 \Rightarrow x = 3, y = -9$: $(3, -9)$

31 Pemberton Ex 10.7 · Q2

$$3x^2 + 6x - 45 = 0 \Rightarrow x^2 + 2x - 15 = 0 \Rightarrow (x + 5)(x - 3) = 0$$

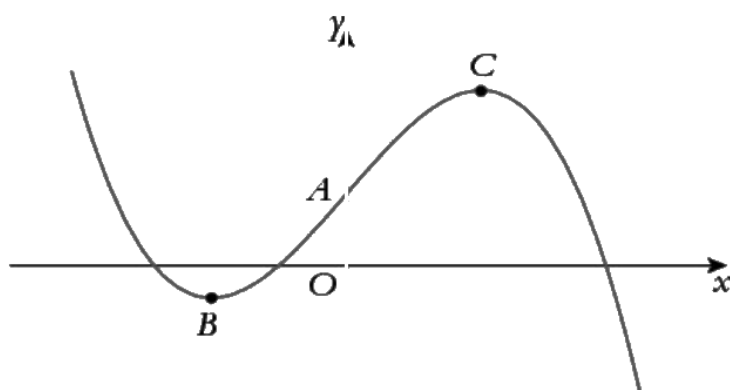
$$A(-5, 175), B(3, -81)$$

32



No calculator

Pemberton Ex 10.7 · Q3



The diagram shows the graph of $y = 11 + 12x - x^3$.

The curve crosses the y -axis at the point A .

It has a minimum point at B and a maximum point at C .

- Find the gradient of the curve at the point A .
- Find the coordinates of the points B and C .

33



No calculator

Pemberton Ex 10.7 · Q4

The curve with equation $y = x^4 - 4x$ has one turning point.
Find the coordinates of this turning point.

34



No calculator

Pemberton Ex 10.7 · Q5

A curve has equation $y = 24x - 3x^2 - x^3$.

Find the coordinates of the turning points on this curve.

32 *Pemberton Ex 10.7 · Q3*

(a) $\frac{dy}{dx} = 12 - 3x^2 = \boxed{12}$ at $x = 0$

(b) $12 - 3x^2 = 0 \Rightarrow x = \pm 2$: $\boxed{B(-2, -5), C(2, 27)}$

33 *Pemberton Ex 10.7 · Q4*

$4x^3 - 4 = 0 \Rightarrow x = 1, y = -3$: $\boxed{(1, -3)}$

34 *Pemberton Ex 10.7 · Q5*

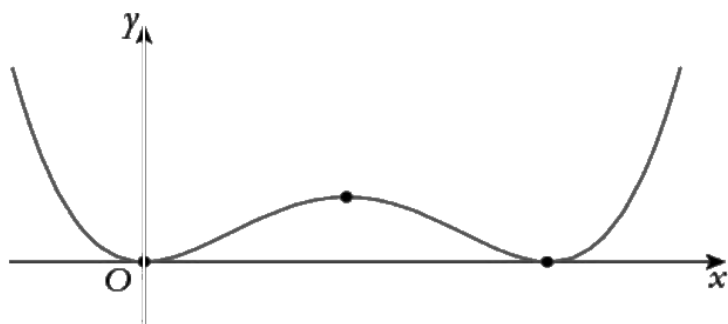
$24 - 6x - 3x^2 = 0 \Rightarrow x^2 + 2x - 8 = 0 \Rightarrow (x + 4)(x - 2) = 0$: $\boxed{(2, 28), (-4, -80)}$

35



No calculator

Pemberton Ex 10.7 · Q6



The diagram shows the graph of $y = x^4 - 4x^3 + 4x^2$.

Use differentiation to find the coordinates of the three turning points.

36



No calculator

Pemberton Ex 10.7 · Q7

Find the coordinates of the stationary points on each of the following curves.

Use graphing software to check your answers

a $y = x^2 - 4x + 8$

b $y = x^3 - 12x + 6$

c $y = x^4 + 4x - 1$

d $y = x(x^2 - 3x - 9)$

e $y = (3 + x)(2 - x)$

f $y = (2x - 3)^3 - 6x$

37



No calculator

Pemberton Ex 10.7 · Q8

The curve $y = x^3 + ax^2 - 9x$ has a minimum point at $x = 1$.

Find the value of a .

35 *Pemberton Ex 10.7 · Q6*

$$4x^3 - 12x^2 + 8x = 4x(x-1)(x-2) = 0: \boxed{(0,0), (1,1), (2,0)}$$

36 *Pemberton Ex 10.7 · Q7*

a $2x - 4 = 0: (2, 4)$

b $3x^2 - 12 = 0: (-2, 22), (2, -10)$

c $4x^3 + 4 = 0: (-1, -4)$

d $3x^2 - 6x - 9 = 0: (-1, 5), (3, -27)$

e $y = 6 - x - x^2; -1 - 2x = 0: (-0.5, 6.25)$

f $6(2x - 3)^2 = 6: (1, -7), (2, -11)$

37 *Pemberton Ex 10.7 · Q8*

$$\frac{dy}{dx} = 3x^2 + 2ax - 9 = 0 \text{ at } x = 1: 3 + 2a - 9 = 0 \Rightarrow \boxed{a = 3}$$

38



No calculator

Pemberton Ex 10.7 · Q9

The curve $y = 2x^3 + ax^2 + bx$ has no turning points.
Show that $a^2 < 6b$.

39



No calculator

Pemberton Ex 10.8 · Q1

The curve with equation $y = x^2 + 2x$ has one turning point

- a Use differentiation to find the coordinates of the turning point.
- b Use the first derivative test to determine whether this point is a maximum or a minimum point.

40



No calculator

Pemberton Ex 10.8 · Q2

The curve with equation $y = 8 + 2x - x^2$ has one turning point

- a Use differentiation to find the coordinates of the turning point.
- b Use the first derivative test to determine whether this point is a maximum or a minimum point.

38 *Pemberton Ex 10.7 · Q9*

$\frac{dy}{dx} = 6x^2 + 2ax + b$. No turning points $\Rightarrow 6x^2 + 2ax + b = 0$ has no real roots:

$$b^2 - 4ac < 0: (2a)^2 - 4(6)(b) < 0 \Rightarrow 4a^2 < 24b \Rightarrow \boxed{a^2 < 6b}$$

39 *Pemberton Ex 10.8 · Q1*

(a) $2x + 2 = 0$: $\boxed{(-1, -1)}$

(b) $x = -1.1$: gradient -0.2 ($-$); $x = -0.9$: $+0.2$ ($+$). $- \ 0 \ + \Rightarrow$
minimum

40 *Pemberton Ex 10.8 · Q2*

(a) $2 - 2x = 0$: $\boxed{(1, 9)}$

(b) $x = 0.9$: $+0.2$; $x = 1.1$: -0.2 . $+ \ 0 \ - \Rightarrow$ **maximum**

41



No calculator

Pemberton Ex 10.8 · Q3

Find the coordinates of the stationary points on each of the following curves and use the first derivative test to determine the nature of each stationary point. Sketch the graph of each function and use graphing software to check your graphs.

a $y = x^2 - 10x + 4$

b $y = (3 + x)(2 - x)$

c $y = 2x^3 - 15x^2$

d $y = 2x^3 - 3x^2 - 36x$

e $y = 4x^3 + 9x^2 - 12x$

f $y = 3x^2 - \frac{2}{3}x^3 - 4x$

41 *Pemberton Ex 10.8 · Q3*

a $2x - 10 = 0 : (5, -21) \text{ min}$

b $-1 - 2x = 0 : (-\frac{1}{2}, 6\frac{1}{4}) \text{ max}$

c $6x(x - 5) = 0 : (0, 0) \text{ max}, (5, -125) \text{ min}$

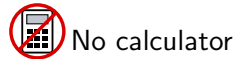
d $6(x - 3)(x + 2) = 0 : (-2, 44) \text{ max}, (3, -81) \text{ min}$

e $6(2x - 1)(x + 2) = 0 : (-2, 28) \text{ max}, (\frac{1}{2}, -3\frac{1}{4}) \text{ min}$

f $-2(x - 1)(x - 2) = 0 : (1, -1\frac{2}{3}) \text{ min}, (2, -1\frac{1}{3}) \text{ max}$

Check each by the gradient sign just either side (+0 – max, –0 + min).

42



Pemberton Ex 10.9 · Q1

Find $\frac{d^2y}{dx^2}$ for each of the following functions.

a $y = x^2 - 4x - 3$

b $y = 2x^3 + 3x^2 - 5x$

c $y = x^4 - x^3$

d $y = 5 - x - 2x^2$

e $y = 1 - 2x + 4x^3$

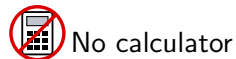
f $y = 2x^5 - 3x^4 + 2$

g $y = x(3 - x^2)$

h $y = (x - 1)(x^2 + 2)$

i $y = 2x^2(5 - 3x + x^2)$

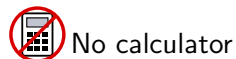
43



Pemberton Ex 10.9 · Q2

Given that $y = 3x^3 - 5x^2 - 1$, find the value of $\frac{d^2y}{dx^2}$ when $x = 2$.

44



Pemberton Ex 10.9 · Q3

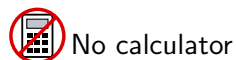
Find $f''(x)$ for each of the following functions.

a $f(x) = 6x^4 - 2x^3$

b $f(x) = 5x^2 - x^5$

c $f(x) = (x - 3)^2 - 5$

45



Pemberton Ex 10.9 · Q4

Given that $f(x) = x^3 - x^2 + 5x$, find

a $f(1)$,

b $f'(1)$,

c $f''(1)$.

42 *Pemberton Ex 10.9 · Q1*

a 2

d -4

g $-6x$

b $12x + 6$

e $24x$

h $6x - 2$

c $12x^2 - 6x$

f $40x^3 - 36x^2$

i $20 - 36x + 24x^2$

43 *Pemberton Ex 10.9 · Q2*

$$\frac{dy}{dx} = 9x^2 - 10x, \quad \frac{d^2y}{dx^2} = 18x - 10 = \boxed{26} \text{ at } x = 2$$

44 *Pemberton Ex 10.9 · Q3*

a $72x^2 - 12x$

b $10 - 20x^3$

c 2

45 *Pemberton Ex 10.9 · Q4*

(a) $f(1) = \boxed{5}$ (b) $f'(x) = 3x^2 - 2x + 5, f'(1) = \boxed{6}$ (c) $f''(x) = 6x - 2,$
 $f''(1) = \boxed{4}$

46



No calculator

Pemberton Ex 10.9 · Q5

Given that $f(x) = (x^2 - 2)^2$, find the value of $f''(-1)$.

47



No calculator

Pemberton Ex 10.9 · Q6

A curve has equation $y = x^3 - 6x^2 + 9x$.

- a Show that $\frac{dy}{dx} = 0$ when $x = 1$ and when $x = 3$.
- b Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$ and when $x = 3$.

48



No calculator

Pemberton Ex 10.9 · Q7

A curve has equation $y = x^3 + ax^2 + bx$. Given that $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = 12$ when $x = 1$, find the value of a and the value of b .

46 *Pemberton Ex 10.9 · Q5*

$$f(x) = x^4 - 4x^2 + 4, \quad f'(x) = 4x^3 - 8x, \quad f''(x) = 12x^2 - 8, \quad f''(-1) = \boxed{4}$$

47 *Pemberton Ex 10.9 · Q6*

$$(a) \frac{dy}{dx} = 3x^2 - 12x + 9 = 3(x-1)(x-3) = 0 \text{ at } x = 1, 3 \checkmark$$

$$(b) \frac{d^2y}{dx^2} = 6x - 12: \quad \boxed{-6} \text{ at } x = 1, \quad \boxed{6} \text{ at } x = 3$$

48 *Pemberton Ex 10.9 · Q7*

$$\frac{d^2y}{dx^2} = 6x + 2a = 12 \text{ at } x = 1 \Rightarrow a = 3. \quad \frac{dy}{dx} = 3 + 2a + b = 0 \Rightarrow b = -9.$$

$$\boxed{a = 3, b = -9}$$

49



No calculator

Pemberton Ex 10.9 · Q8

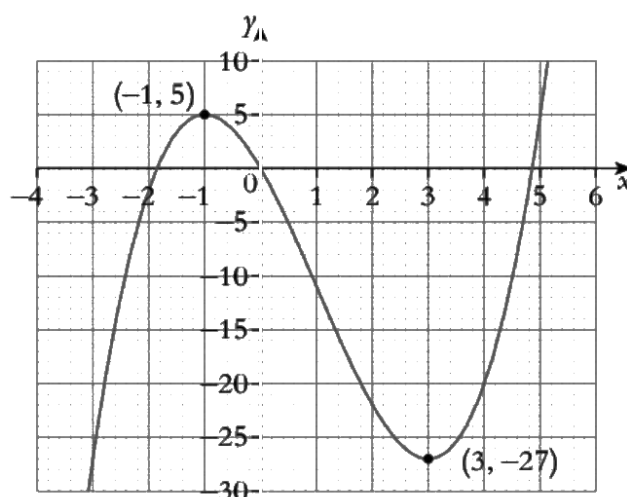
A curve has equation $y = x^3 - 3x^2 - 9x$.

a Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

b Copy and complete the table to show whether $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ are positive (+), negative (−) or zero (0) for the given values of x .

x	−3	−2	−1	0	1	2	3	4	5
$\frac{dy}{dx}$						−			
$\frac{d^2y}{dx^2}$						+			

c The graph of $y = x^3 - 3x^2 - 9x$ is shown below, together with the coordinates of its maximum and minimum points.



Use the word *positive* or the word *negative* to complete each of these statements.

- i At the maximum point $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2}$ is
- ii At the minimum point $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2}$ is

49 *Pemberton Ex 10.9 · Q8*

(a) $\frac{dy}{dx} = \boxed{3x^2 - 6x - 9}, \quad \frac{d^2y}{dx^2} = \boxed{6x - 6}$

(b)

x	-3	-2	-1	0	1	2	3	4	5
$\frac{dy}{dx}$	+	+	0	-	-	-	0	+	+
$\frac{d^2y}{dx^2}$	-	-	-	-	0	+	+	+	+

(c) (i) **negative** (ii) **positive**

50



No calculator

Pemberton Ex 10.10 · Q1

The curve with equation $y = 10x - x^2$ has one turning point

- a Use differentiation to find the coordinates of the turning point.
- b Use the second derivative test to determine whether this point is a maximum or a minimum point.

51



No calculator

Pemberton Ex 10.10 · Q2

The curve with equation $y = x^2 - 2x - 8$ has one turning point

- a Use differentiation to find the coordinates of the turning point.
- b Use the second derivative test to determine whether this point is a maximum or a minimum point.

52



No calculator

Pemberton Ex 10.10 · Q3

Find the coordinates of the stationary points on each of the following curves and use the second derivative test to determine the nature of each stationary point. Sketch the graph of each function and use graphing software to check your graphs.

- | | |
|-----------------------|---------------------------|
| a $y = 2x^2 + 8x + 5$ | b $y = 1 - (2x + 1)^2$ |
| c $y = x^4 + 4x - 1$ | d $y = 4x^3 - 3x^2 - 36x$ |
| e $y = x^3 - 12x + 6$ | f $y = 9x^2 - 15x - x^3$ |

50 *Pemberton Ex 10.10 · Q1*

(a) $10 - 2x = 0$: $\boxed{(5, 25)}$ (b) $\frac{d^2y}{dx^2} = -2 < 0 \Rightarrow$ **maximum**

51 *Pemberton Ex 10.10 · Q2*

(a) $2x - 2 = 0$: $\boxed{(1, -9)}$ (b) $\frac{d^2y}{dx^2} = 2 > 0 \Rightarrow$ **minimum**

52 *Pemberton Ex 10.10 · Q3*

a $4x + 8 = 0$: $(-2, -3)$; $y = 4$, min

b $-4(2x + 1) = 0$: $(-\frac{1}{2}, 1)$; $y = -8$, max

c $4x^3 + 4 = 0$: $(-1, -4)$; $y = 12$, min

d $(-1.5, 33.75)$ max; $(2, -52)$ min

e $(-2, 22)$ max; $(2, -10)$ min

f $(1, -7)$ min; $(5, 25)$ max

(d) $12x^2 - 6x - 36 = 0 \Rightarrow (2x + 3)(x - 2) = 0$, $y'' = 24x - 6$. (f) $-3x^2 + 18x - 15 = 0 \Rightarrow x = 1, 5$, $y'' = 18 - 6x$.

53

No calculator

Pemberton Ex 10.10 · Q4

The curve $y = 2x^3 + ax^2 + bx$ has a stationary point when $x = 2$.

The curve passes through the point $(1, 23)$.

- a Find the value of a and the value of b .
- b Find the coordinates of the other stationary point on the curve and determine the nature of this point.

54

No calculator

Pemberton Ex 10.10 · Q5

The curve $y = x^3 + ax^2 + b$ has a stationary point at the point $(4, -27)$.

- a Find the value of a and the value of b .
- b Determine the nature of the stationary point $(4, -27)$.
- c Find the coordinates of the other stationary point on the curve and determine the nature of this stationary point.

55

No calculator

Pemberton Ex 10.10 · Q6

The curve $y = 2x^3 + ax^2 + bx$ has a stationary point at the point $(-1, 7)$.

- a Find the value of a and the value of b .
- b Find the coordinates of the second stationary point on the curve.
- c Determine the nature of the two stationary points.

53 Pemberton Ex 10.10 · Q4

(a) $\frac{dy}{dx} = 6x^2 + 2ax + b$: $24 + 4a + b = 0$; point: $2 + a + b = 23$

Subtract: $22 + 3a = -23 \Rightarrow \boxed{a = -15, b = 36}$

(b) $6x^2 - 30x + 36 = 0 \Rightarrow x = 2, 3$. $\boxed{(3, 27)}$, $y'' = 12x - 30 = 6 > 0$

minimum

54 Pemberton Ex 10.10 · Q5

(a) $3x^2 + 2ax = 0$ at $x = 4$: $48 + 8a = 0 \Rightarrow a = -6$; $64 - 96 + b = -27 \Rightarrow$

$\boxed{a = -6, b = 5}$

(b) $y'' = 6x - 12 = 12 > 0$ **minimum**

(c) $3x^2 - 12x = 0 \Rightarrow x = 0$: $\boxed{(0, 5)}$, $y'' = -12 < 0$ **maximum**

55 Pemberton Ex 10.10 · Q6

(a) $6 - 2a + b = 0$ and $-2 + a - b = 7 \Rightarrow \boxed{a = -3, b = -12}$

(b) $6x^2 - 6x - 12 = 0 \Rightarrow x = -1, 2$: $\boxed{(2, -20)}$

(c) $y'' = 12x - 6$: $(-1, 7)$ **maximum**, $(2, -20)$ **minimum**

56

No calculator

Oct/Nov 2023 · Paper 22 · Q25

The derivative of $2ax^7 + 3x^k$ is $42x^6 + 15x^{k-1}$.

Find the value of a and the value of k .

$a = \dots\dots\dots$

$k = \dots\dots\dots$ [2]

57

No calculator

May/June 2023 · Paper 21 · Q21

A curve has equation $y = x^3 - 12x$.

(a) Find the gradient of the curve at the point $(1, -11)$.

$\dots\dots\dots$ [3]

(b) Find the coordinates of the turning points of the curve.

$(\dots\dots\dots, \dots\dots\dots)$ and $(\dots\dots\dots, \dots\dots\dots)$ [3]

58

No calculator

May/June 2021 · Paper 23 · Q25

Find the x -coordinates of the points on the graph of $y = x^5 - 5x^4$ where the gradient is 0.

$\dots\dots\dots$ [4]

56 Oct/Nov 2023 · Paper 22 · Q25

$$\frac{d}{dx}(2ax^7 + 3x^k) = 14ax^6 + 3kx^{k-1}. \quad 14a = 42 \Rightarrow \boxed{a = 3}; \quad 3k = 15 \Rightarrow \boxed{k = 5}$$

57 May/June 2023 · Paper 21 · Q21

$$\begin{aligned} \text{(a)} \quad \frac{dy}{dx} &= 3x^2 - 12 = 3 - 12 = \boxed{-9} & \text{(b)} \quad 3x^2 - 12 = 0 \Rightarrow x = \pm 2: \\ & \boxed{(2, -16) \text{ and } (-2, 16)} \end{aligned}$$

58 May/June 2021 · Paper 23 · Q25

$$\frac{dy}{dx} = 5x^4 - 20x^3 = 5x^3(x - 4) = 0 \Rightarrow \boxed{x = 0 \text{ and } x = 4}$$

59

No calculator

Oct/Nov 2020 • Paper 22 • Q21

(a) Differentiate $6 + 4x - x^2$.

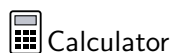
..... [2]

(b) Find the coordinates of the turning point of the graph of $y = 6 + 4x - x^2$.

(..... ,) [2]

59 *Oct/Nov 2020 · Paper 22 · Q21*

(a) $4 - 2x$ (b) $4 - 2x = 0 \Rightarrow x = 2, y = 6 + 8 - 4 = 10$: $(2, 10)$

60**Oct/Nov 2024 · Paper 42 · Q10**

A curve has the equation $y = x^3 - 9x^2 - 48x$.

- (a) Differentiate $x^3 - 9x^2 - 48x$.

..... [2]

- (b) Find the coordinates of the turning points of the graph of $y = x^3 - 9x^2 - 48x$.
You must show all your working.

(..... ,) and (..... ,)
[4]

- (c) Determine whether each of the turning points is a maximum or a minimum.
Give reasons for your answers.

[3]

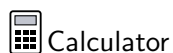
60 Oct/Nov 2024 · Paper 42 · Q10

(a) $3x^2 - 18x - 48$ (b) $3(x - 8)(x + 2) = 0$: $(8, -448)$ and $(-2, 52)$

(c) $\frac{d^2y}{dx^2} = 6x - 18$: at $x = 8$: $30 > 0$ **minimum**; at $x = -2$: $-30 < 0$

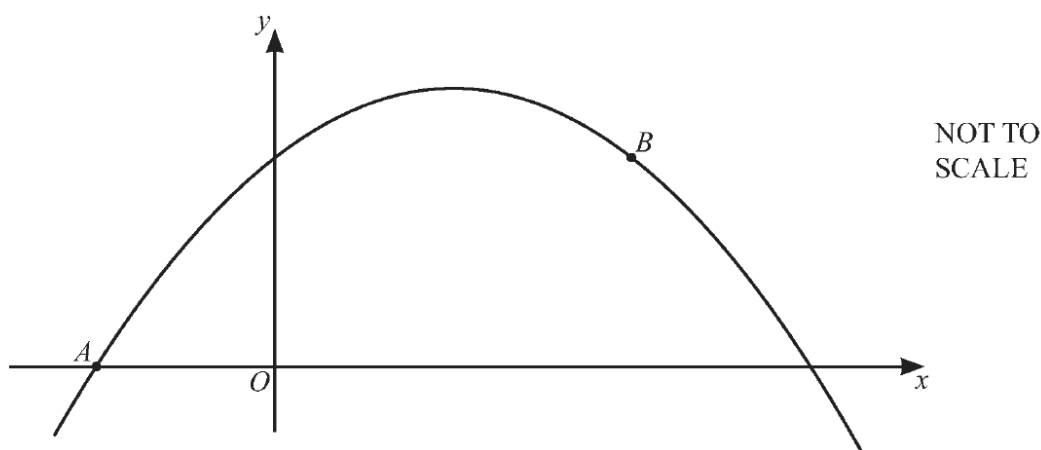
maximum

61



Calculator

Oct/Nov 2024 · Paper 43 · Q10



The diagram shows a sketch of the graph of $y = 3 + 2x - x^2$.
 A is the point $(-1, 0)$ and B is the point $(2, 3)$.

- (a) Find the derivative of $3 + 2x - x^2$.

..... [2]

- (b) (i) Show that the equation of the tangent at A is $y = 4x + 4$.

[3]

- (ii) The line L is perpendicular to the line $y = 4x + 4$.
 The line L passes through the point B .

Find the equation of the line L .

Give your answer in the form $y = mx + c$.

$y =$ [3]

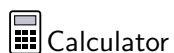
- (c) Find the coordinates of the maximum point on the graph of $y = 3 + 2x - x^2$.

(..... ,) [3]

61 *Oct/Nov 2024 · Paper 43 · Q10*

- (a) $2 - 2x$ (b)(i) gradient at A: $2 + 2 = 4$; $y = 4(x + 1) = 4x + 4$ ✓
(ii) gradient $-\frac{1}{4}$ through $(2, 3)$: $y = -\frac{1}{4}x + 3.5$ (c) $2 - 2x = 0$: $(1, 4)$

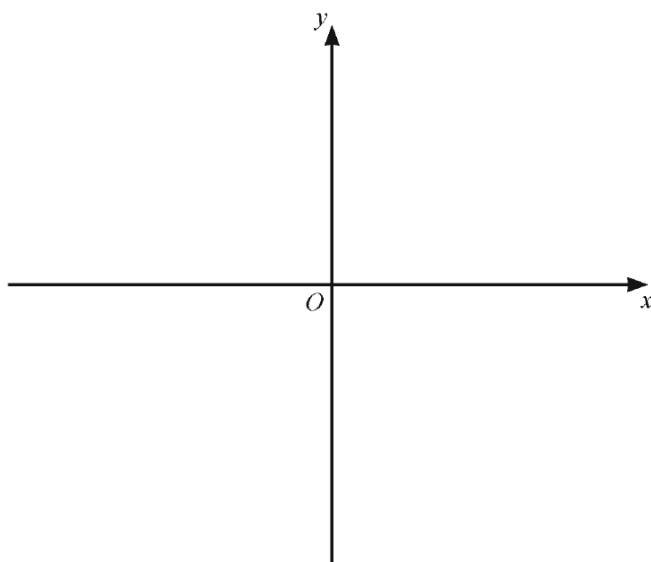
62



Calculator

May/June 2024 • Paper 41 • Q8

- (a) On the axes, sketch the graph of $y = x^2 + 7x - 18$.
On your sketch, write the values where the graph meets the x -axis and the y -axis.



[4]

- (b) (i) Find the derivative of $y = x^2 - 3x - 28$.

..... [2]

- (ii) Find the coordinates of the turning point of $y = x^2 - 3x - 28$.

(..... ,) [3]

- (c) The line $y = 5 - 2x$ intersects the graph of $y = x^2 - 3x - 28$ at point P and point Q .

Find the coordinates of P and Q .

You must show all your working and give your answers correct to 2 decimal places.

(..... ,)

(..... ,) [6]

62 May/June 2024 · Paper 41 · Q8

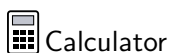
(a) $(x+9)(x-2)$: U-shape through $(-9, 0)$, $(2, 0)$, $(0, -18)$

(b)(i) $2x - 3$ (ii) $x = 1.5$: $(1.5, -30.25)$

(c) $x^2 - 3x - 28 = 5 - 2x \Rightarrow x^2 - x - 33 = 0 \Rightarrow x = \frac{1 \pm \sqrt{133}}{2}$

$(6.27, -7.53)$ and $(-5.27, 15.53)$

63



May/June 2024 · Paper 42 · Q11

- (a) The point $(-1, 6)$ lies on a curve.

This curve has the derived function $\frac{dy}{dx} = -4x^3 - 9x^2 + 5$.

Show that $(-1, 6)$ is a stationary point of the curve.

[2]

- (b) A different curve has equation $y = 2x^3 - 6x + 8$.

- (i) Calculate the gradient of the tangent to this curve at the point $(-2, 2)$.

..... [3]

- (ii) Find the x -coordinates of the stationary points of this curve.

 $x = \dots\dots\dots$ and $x = \dots\dots\dots$ [2]

63 *May/June 2024 · Paper 42 · Q11*

(a) At $x = -1$: $-4(-1)^3 - 9(-1)^2 + 5 = 4 - 9 + 5 = 0$, so $(-1, 6)$ is stationary ✓

(b)(i) $\frac{dy}{dx} = 6x^2 - 6 = \boxed{18}$ (ii) $6x^2 - 6 = 0$: $\boxed{x = 1 \text{ and } x = -1}$

64

 Calculator

May/June 2024 · Paper 43 · Q10

$$y = x^7 - 7x^6$$

- (a) Find the derivative of y with respect to x .

..... [2]

- (b) Find the equation of the tangent to the graph of $y = x^7 - 7x^6$ at the point where $x = -1$.
Give your answer in the form $y = mx + c$.

$y =$ [4]

- (c) The graph of $y = x^7 - 7x^6$ has two turning points.

Find the coordinates of these points.

You must show all your working.

(..... ,)

(..... ,) [5]

64 *May/June 2024 · Paper 43 · Q10*

(a) $7x^6 - 42x^5$ (b) at $x = -1$: gradient $7 + 42 = 49$, $y = -1 - 7 = -8$:

$$y = 49x + 41$$

(c) $7x^5(x - 6) = 0$: $(0, 0)$ and $(6, -46656)$

65

Calculator

Oct/Nov 2023 · Paper 41 · Q11

- (a) Differentiate $x^3 - 4x^2 - 3x$.

..... [2]

- (b) A curve has equation $y = x^3 - 4x^2 - 3x$.

Work out the coordinates of the two stationary points.
Show all your working.

(..... ,)

(..... ,) [5]

- (c) Determine whether each stationary point is a maximum or a minimum.
Show all your working.

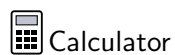
[3]

65 Oct/Nov 2023 · Paper 41 · Q11

(a) $3x^2 - 8x - 3$ (b) $(3x + 1)(x - 3) = 0$: $(3, -18)$ and $(-\frac{1}{3}, \frac{14}{27})$

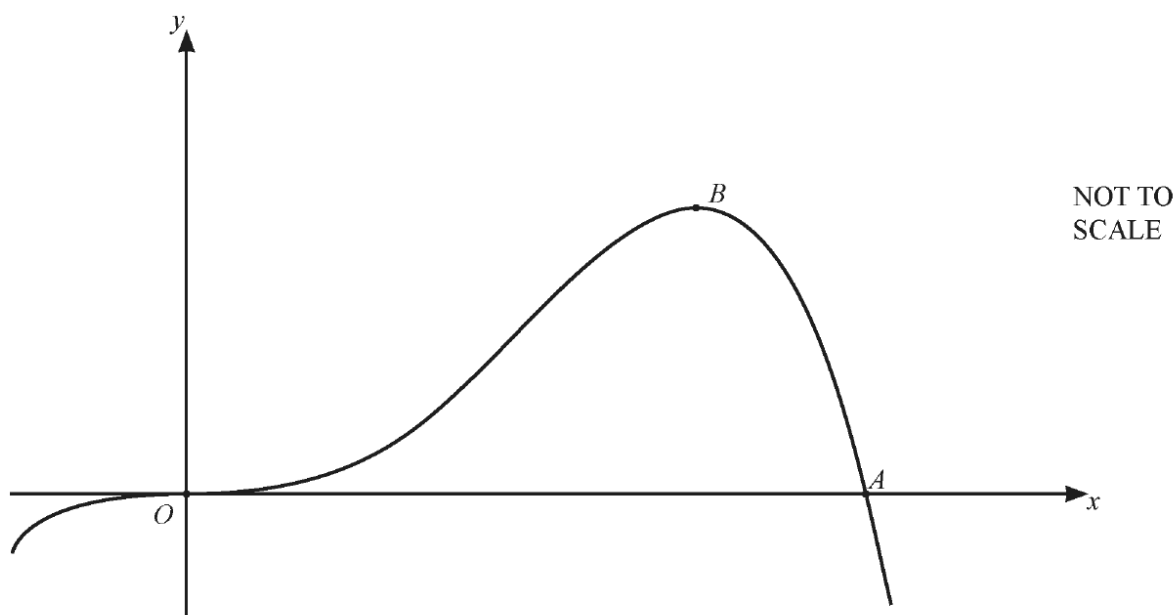
(c) $\frac{d^2y}{dx^2} = 6x - 8$: at $x = 3$: $10 > 0$ **minimum**; at $x = -\frac{1}{3}$: $-10 < 0$
maximum

66



Calculator

Oct/Nov 2023 · Paper 42 · Q9



The diagram shows a sketch of the graph of $y = 4x^3 - x^4$.
 The graph crosses the x -axis at the origin O and at the point A .
 The point B is a maximum point.

(a) Differentiate $4x^3 - x^4$.

..... [2]

(b) Find the coordinates of B .

(.....,) [3]

(c) Find the gradient of the graph at the point A .

..... [3]

66 Oct/Nov 2023 · Paper 42 · Q9

(a) $12x^2 - 4x^3$ (b) $4x^2(3 - x) = 0 \Rightarrow x = 3, y = 108 - 81$: $B(3, 27)$

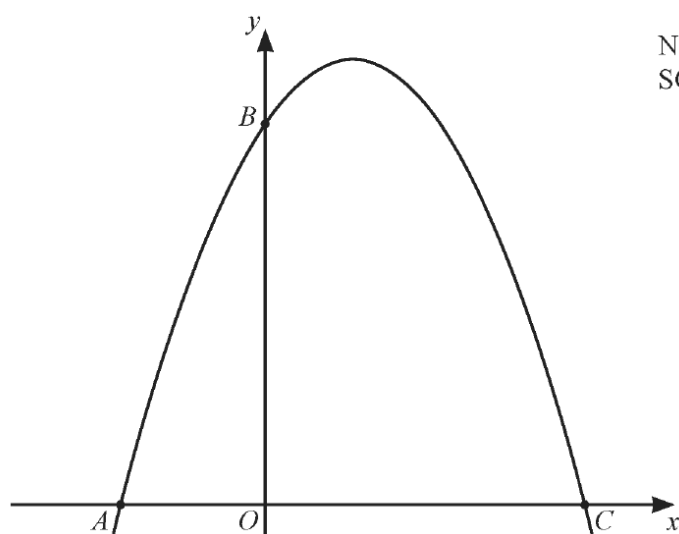
(c) A: $x^3(4 - x) = 0 \Rightarrow x = 4$; gradient = $192 - 256 = -64$

67



Calculator

Oct/Nov 2023 · Paper 43 · Q11

NOT TO
SCALE

The diagram shows a sketch of $y = 18 + 5x - 2x^2$.

(a) Find the coordinates of the points A , B and C .

A (..... ,)

B (..... ,)

C (..... ,) [4]

(b) Differentiate $18 + 5x - 2x^2$.

..... [2]

(c) Find the coordinates of the point on $y = 18 + 5x - 2x^2$ where the gradient is 17.

(..... ,) [3]

67 *Oct/Nov 2023 · Paper 43 · Q11*

(a) $-(2x - 9)(x + 2) = 0$: $A(-2, 0), B(0, 18), C(4.5, 0)$ (b) $5 - 4x$

(c) $5 - 4x = 17 \Rightarrow x = -3, y = 18 - 15 - 18$: $(-3, -15)$

68



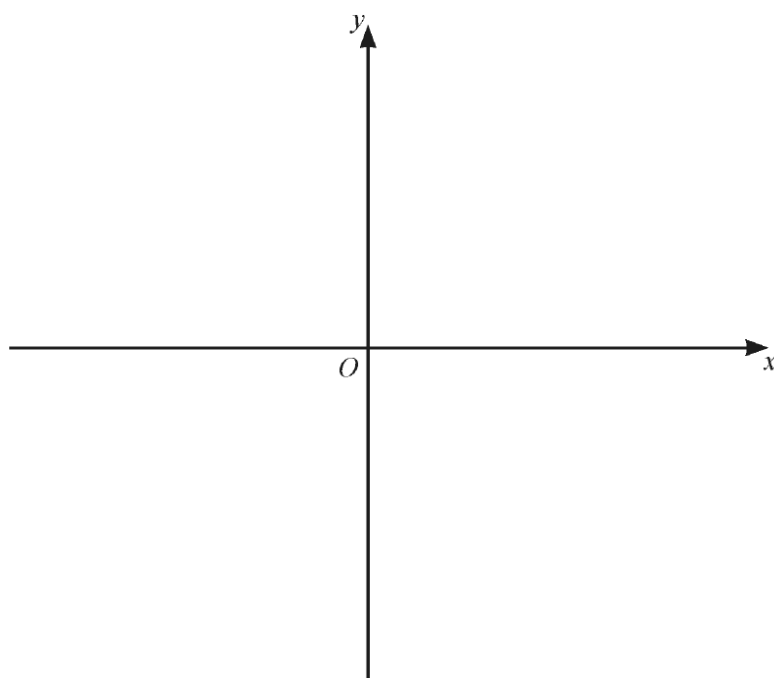
Calculator

May/June 2023 · Paper 42 · Q8

- (a) (i) Show that the equation $y = (x-4)(x+1)(x-2)$ can be written as $y = x^3 - 5x^2 + 2x + 8$.

[2]

- (ii) On the diagram, sketch the graph of $y = x^3 - 5x^2 + 2x + 8$, indicating the values where the graph crosses the axes.



[4]

- (b) The graph of $y = x^3 - 5x^2 + 2x + 8$ has two tangents with a gradient of 10.

Find the equations of these two tangents.

You must show all your working and give your answers in the form $y = mx + c$.

$y = \dots\dots\dots$

$y = \dots\dots\dots$ [7]

68 *May/June 2023 · Paper 42 · Q8*

(a)(i) $(x - 4)(x^2 - x - 2) = x^3 - 5x^2 + 2x + 8$ ✓ (ii) cubic crossing at $-1, 2, 4$; y-intercept 8

(b) $3x^2 - 10x + 2 = 10 \Rightarrow (3x + 2)(x - 4) = 0$

$x = 4$: $y = 0$: $y = 10x - 40$; $x = -\frac{2}{3}$: $y = \frac{112}{27}$: $y = 10x + \frac{292}{27}$ ($\approx 10x + 10.8$)

69 Calculator**May/June 2023 · Paper 43 · Q12**

The equation of a curve is $y = x^4 - 8x^2 + 5$.

- (a) Find the derivative, $\left(\frac{dy}{dx}\right)$, of $y = x^4 - 8x^2 + 5$.

..... [2]

- (b) Find the coordinates of the three turning points.
You must show all your working.

(..... ,) and (..... ,) and (..... ,) [4]

- (c) Determine which one of these turning points is a maximum.
Justify your answer.

[2]

69 *May/June 2023 · Paper 43 · Q12*

(a) $4x^3 - 16x$ (b) $4x(x^2 - 4) = 0$: $(0, 5), (2, -11), (-2, -11)$

(c) $\frac{d^2y}{dx^2} = 12x^2 - 16 = -16 < 0$ at $x = 0$: $(0, 5)$ is the **maximum**

70

Calculator

Oct/Nov 2022 · Paper 42 · Q6**(a)** Solve.

$$4x + 15 = 9$$

$$x = \dots\dots\dots [2]$$

(b) Factorise.

$$a^2 - 9$$

$$\dots\dots\dots [1]$$

(c) Write as a single fraction in its simplest form.

$$\frac{4a}{5} \div \frac{3ad}{10c}$$

$$\dots\dots\dots [3]$$

(d) $5^n + 5^n + 5^n + 5^n + 5^n = 5^m$

Find an expression for m in terms of n .

$$m = \dots\dots\dots [2]$$

(e) Solve by factorisation.

$$4x^2 + 8x - 5 = 0$$

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots [3]$$

(f) (i) y is directly proportional to $(x+3)^3$.
When $x = 2$, $y = 13.5$.Find x when $y = 108$.*Question continues on the next page →*

$$x = \dots\dots\dots [3]$$

- (ii) g is inversely proportional to the square of d .
When d is halved, the value of g is multiplied by a factor n .

Find n .

$$n = \dots\dots\dots [2]$$

- (g) Expand and simplify.

$$(2x + 3)(x - 1)(x + 3)$$

$$\dots\dots\dots [3]$$

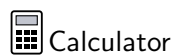
- (h) Find the derivative, $\frac{dy}{dx}$, of $y = 3x^2 + 4x - 1$.

$$\dots\dots\dots [2]$$

70 Oct/Nov 2022 · Paper 42 · Q6

- (a) -1.5 (b) $(a-3)(a+3)$ (c) $\frac{4a}{5} \times \frac{10c}{3ad} = \frac{8c}{3d}$ (d) 5×5^n :
 $m = n + 1$
- (e) $(2x-1)(2x+5) = 0$: $x = 0.5$ or -2.5 (f)(i) $y = 0.108(x+3)^3$,
 $(x+3)^3 = 1000$: $x = 7$ (ii) $n = 4$
- (g) $2x^3 + 7x^2 - 9$ (h) $6x + 4$

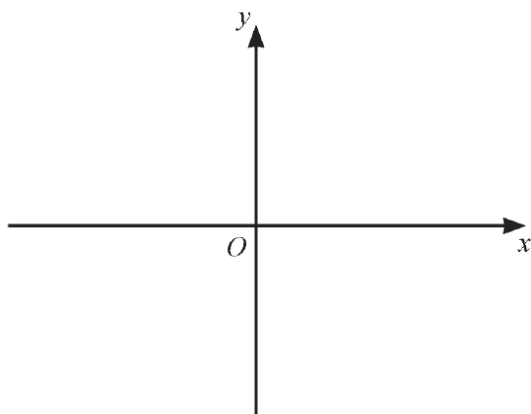
71



Oct/Nov 2022 · Paper 43 · Q9

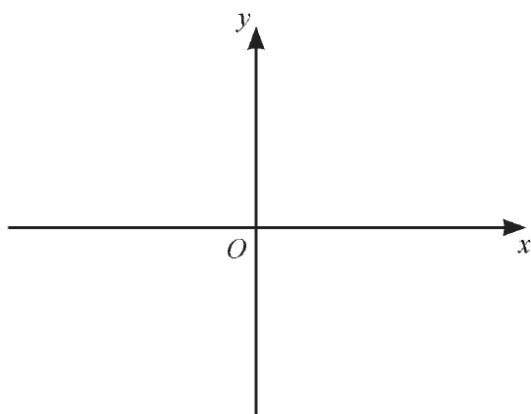
- (a) Sketch the following graphs.
On each sketch, indicate any intercepts with the axes.

(i) $3x - 4y = 12$



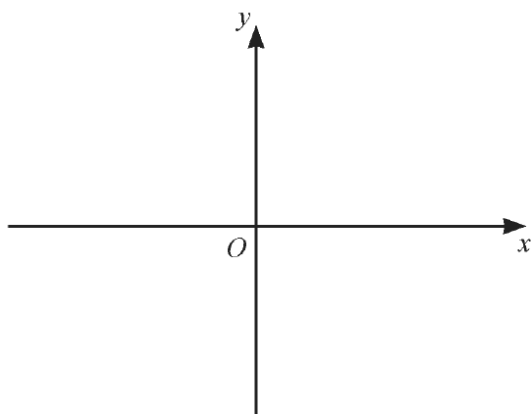
[2]

(ii) $y = x^2 - 3x - 4$



[4]

(iii) $y = 6^x$



[2]

Question continues on the next page →

- (b) (i) Find the derivative, $\frac{dy}{dx}$, of $y = 5 + 8x - \frac{4}{3}x^3$.

..... [2]

- (ii) Find the gradient of $y = 5 + 8x - \frac{4}{3}x^3$ at $x = -1$.

..... [2]

- (iii) A tangent is drawn to the graph of $y = 5 + 8x - \frac{4}{3}x^3$.

The gradient of the tangent is -28 .

Find the coordinates of the two possible points where this tangent meets the graph.

(..... ,)

(..... ,) [5]

71 Oct/Nov 2022 · Paper 43 · Q9

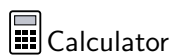
(a)(i) line through $(4, 0)$ and $(0, -3)$ (ii) U-shape through $(-1, 0)$, $(4, 0)$, $(0, -4)$ (iii) exponential through $(0, 1)$, above the x -axis

(b)(i) $8 - 4x^2$ (ii) $8 - 4 = 4$

(iii) $8 - 4x^2 = -28 \Rightarrow x = \pm 3$: $(3, -7)$ and $(-3, 17)$

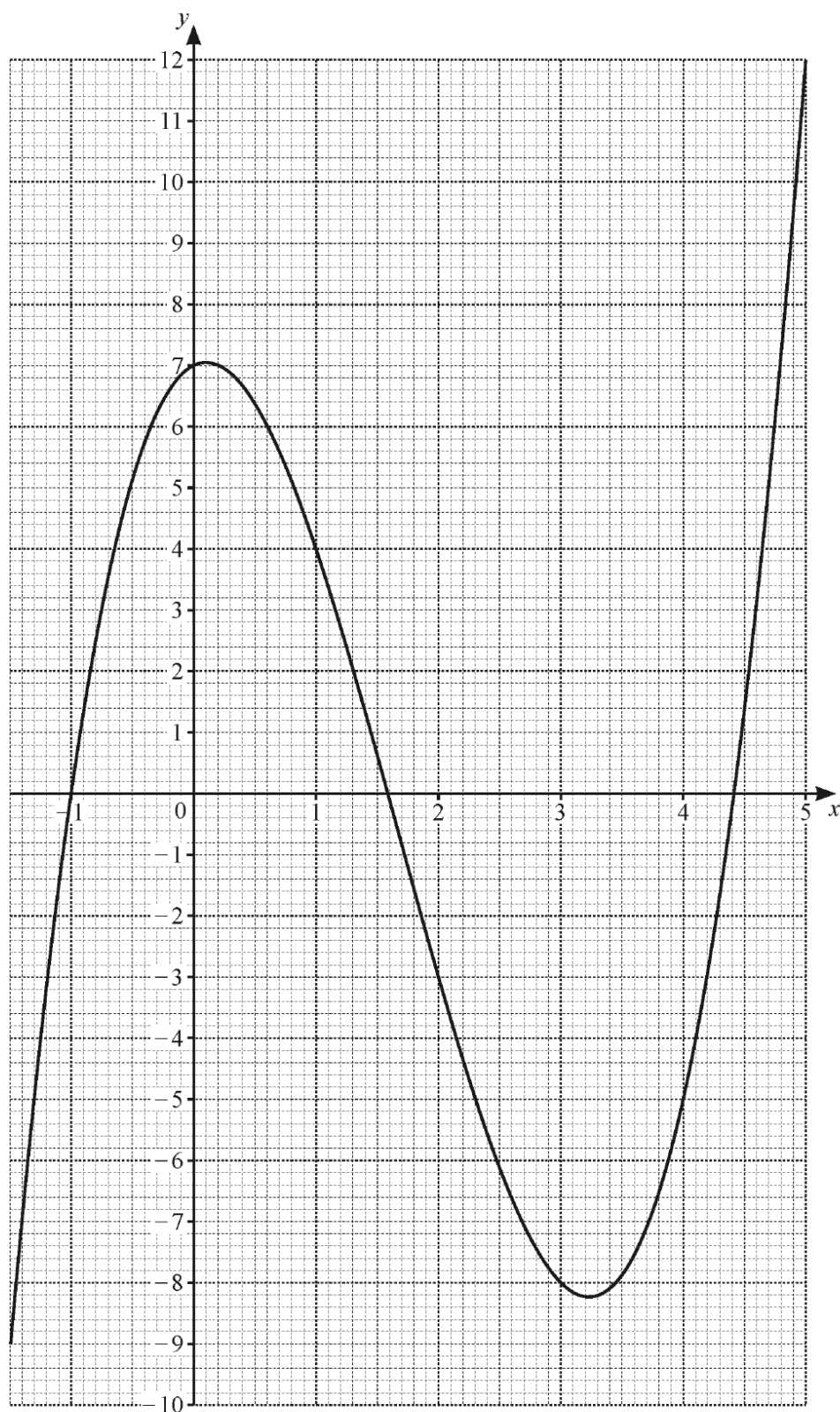
72

(a)



Calculator

May/June 2022 · Paper 41 · Q6



Question continues on the next page →

The diagram shows the graph of $y = f(x)$ for $-1.5 \leq x \leq 5$.

(i) Find $f(2)$.

..... [1]

(ii) Solve the equation $f(x) = 0$ for $-1.5 \leq x \leq 5$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

(iii) $f(x) = k$ has three solutions for $-1.5 \leq x \leq 5$ where k is an integer.

Find the smallest possible value of k .

$k = \dots\dots\dots$ [1]

(iv) On the grid, draw a line $y = mx$ so that $f(x) = mx$ has exactly one solution for $-1.5 \leq x \leq 5$.
[2]

(b) $y = 3x^2 - 12x + 7$

(i) Find the value of $\frac{dy}{dx}$ when $x = 5$.

..... [3]

(ii) Find the coordinates of the point on the graph of $y = 3x^2 - 12x + 7$ where the gradient is 0.

(..... ,) [2]

(c) When $y = 2x^p + qx^2$, $\frac{dy}{dx} = 14x^6 + 6x$.

Find the value of p and the value of q .

$p = \dots\dots\dots$

$q = \dots\dots\dots$ [2]

72 May/June 2022 · Paper 41 · Q6

(The curve is $f(x) = x^3 - 5x^2 + x + 7$.)

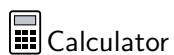
(a)(i) $\boxed{-3}$ (ii) $\boxed{x = -1, 1.6, 4.4}$ (iii) $\boxed{k = -8}$

(iv) e.g. $\boxed{y = -5x}$ or $y = 10x$ (a line through O that meets the curve only once)

(b)(i) $\frac{dy}{dx} = 6x - 12 = \boxed{18}$ (ii) $6x - 12 = 0$: $\boxed{(2, -5)}$

(c) $2p = 14$: $\boxed{p = 7}$; $2q = 6$: $\boxed{q = 3}$

73



Calculator

May/June 2022 · Paper 42 · Q12

A curve has equation $y = x^3 - kx^2 + 1$.
When $x = 2$, the gradient of the curve is 6.

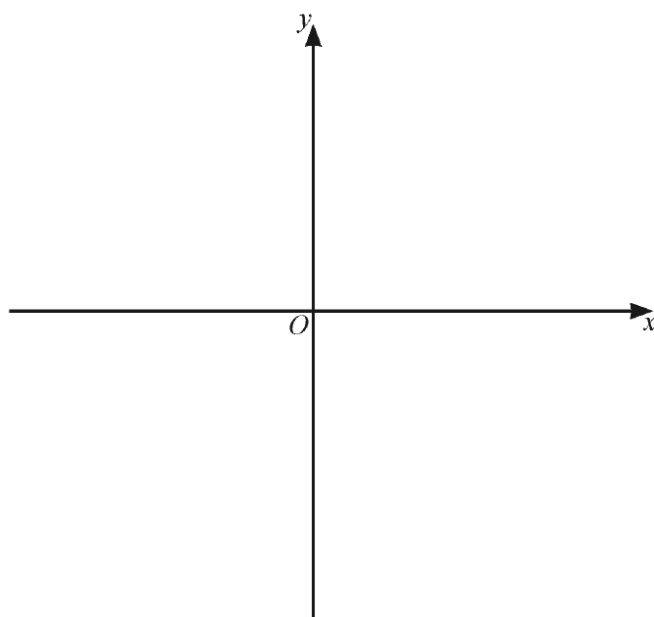
- (a) Show that $k = 1.5$.

[5]

- (b) Find the coordinates of the two stationary points of $y = x^3 - 1.5x^2 + 1$.
You must show all your working.

(.....,) and (.....,) [4]

- (c) Sketch the curve $y = x^3 - 1.5x^2 + 1$.



[2]

73 *May/June 2022 · Paper 42 · Q12*

(a) $\frac{dy}{dx} = 3x^2 - 2kx = 12 - 4k = 6 \Rightarrow k = 1.5 \checkmark$

(b) $3x^2 - 3x = 3x(x - 1) = 0$: $\boxed{(0, 1) \text{ and } (1, 0.5)}$

(c) Cubic: maximum at $(0, 1)$, minimum at $(1, 0.5)$, crosses the x -axis once (at a negative x)

74



Calculator

Oct/Nov 2021 • Paper 41 • Q8

(a) $f(x) = 3 - 5x$

(i) Find x when $f(x) = -5$.

$x = \dots\dots\dots$ [2]

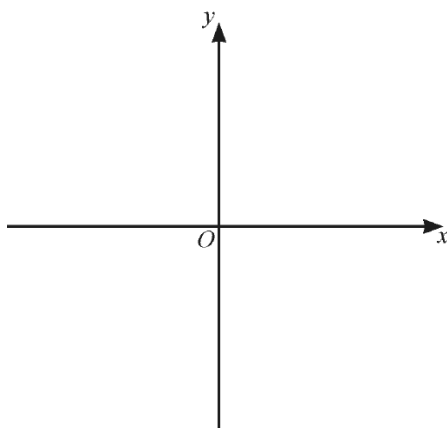
(ii) Find $f^{-1}(x)$.

$f^{-1}(x) = \dots\dots\dots$ [2]

(b) $g(x) = 18 - 3x - x^2$

(i) Write $g(x)$ in the form $b - (a + x)^2$.

$\dots\dots\dots$ [3]

(ii) Sketch the graph of $y = g(x)$.
On your sketch, show the coordinates of the turning point.

[3]

(iii) Find the equation of the tangent to the graph of $y = 18 - 3x - x^2$ at $x = 4$.
Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$ [6]

74 Oct/Nov 2021 · Paper 41 · Q8

(a)(i) $3 - 5x = -5 \Rightarrow x = 1.6$ (ii) $\frac{3-x}{5}$

(b)(i) $18 - (x^2 + 3x) = 18 - [(x + 1.5)^2 - 2.25] = 20.25 - (1.5 + x)^2$

(ii) $\cap$ -shaped, maximum $(-1.5, 20.25)$, y-intercept 18, roots -6 and 3

(iii) $\frac{dy}{dx} = -3 - 2x = -11$ at $x = 4$; $y = -10$: $y = -11x + 34$

75



Calculator

Oct/Nov 2021 · Paper 42 · Q10

- (a) Find the coordinates of the turning points of the graph of $y = x^3 - 12x + 6$.
You must show all your working.

(.....,) and (.....,) [5]

- (b) Determine whether each turning point is a maximum or a minimum.
Show how you decide.

[3]

75 *Oct/Nov 2021 · Paper 42 · Q10*

(a) $3x^2 - 12 = 0 \Rightarrow x = \pm 2$: $\boxed{(2, -10) \text{ and } (-2, 22)}$

(b) $\frac{d^2y}{dx^2} = 6x$: at $x = 2$: $12 > 0$ **minimum**; at $x = -2$: $-12 < 0$ **maximum**

76



Calculator

Oct/Nov 2021 · Paper 43 · Q9

$$f(x) = x(x-1)(x-2)$$

- (a) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x -axis.

(..... ,)

(..... ,)

(..... ,) [2]

- (b) Show that $f(x) = x^3 - 3x^2 + 2x$.

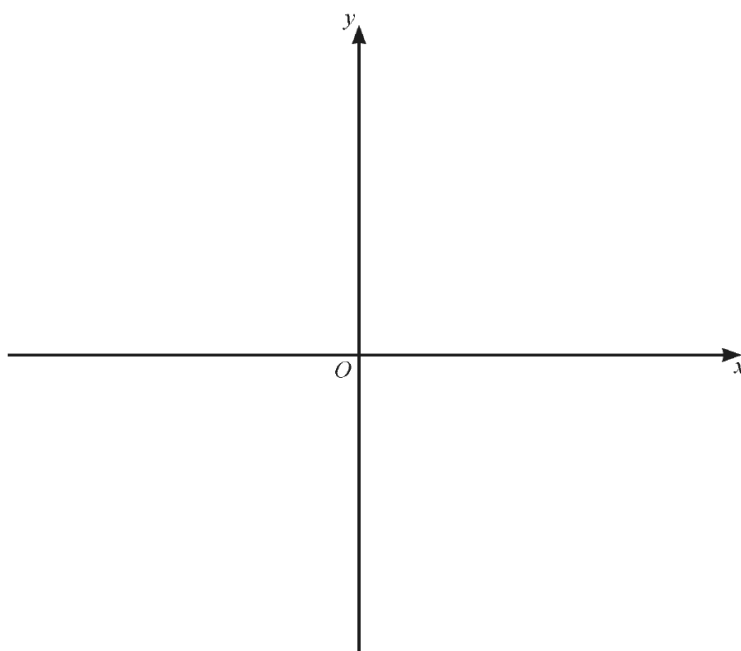
[2]

- (c) Find the coordinates of the turning points of the graph of $y = f(x)$.
Show all your working and give your answers correct to 1 decimal place.

(..... ,)

(..... ,) [8]

- (d) Sketch the graph of $y = f(x)$.



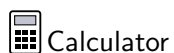
[2]

76 Oct/Nov 2021 · Paper 43 · Q9

(a) $\boxed{(0, 0), (1, 0), (2, 0)}$ (b) $x(x^2 - 3x + 2) = x^3 - 3x^2 + 2x$ ✓

(c) $3x^2 - 6x + 2 = 0 \Rightarrow x = \frac{6 \pm \sqrt{12}}{6} = 1.577 \text{ or } 0.423$: $\boxed{(0.4, 0.4)}$ and $\boxed{(1.6, -0.4)}$

(d) Cubic through 0, 1, 2 on the x -axis: maximum near (0.4, 0.4), minimum near (1.6, -0.4)

77

Calculator

May/June 2021 • Paper 41 • Q7**(a)** Simplify.

$$\frac{x^2 - 25}{x^2 - x - 20}$$

..... [3]

(b) Write as a single fraction in its simplest form.

$$\frac{x+5}{x} + \frac{x+8}{x-1}$$

..... [3]

(c) A curve has equation $y = 2x^3 - 4x^2 + 6$.**(i)** Find $\frac{dy}{dx}$, the derived function of y .

..... [2]

(ii) Calculate the gradient of the curve $y = 2x^3 - 4x^2 + 6$ at $x = 4$.

..... [2]

(iii) Find the coordinates of the two stationary points on the curve.

(.....,) and (.....,) [4]

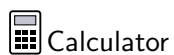
77 May/June 2021 · Paper 41 · Q7

$$(a) \frac{(x-5)(x+5)}{(x-5)(x+4)} = \boxed{\frac{x+5}{x+4}} \quad (b) \frac{(x+5)(x-1) + x(x+8)}{x(x-1)} = \boxed{\frac{2x^2 + 12x - 5}{x(x-1)}}$$

$$(c)(i) \boxed{6x^2 - 8x} \quad (ii) 96 - 32 = \boxed{64}$$

$$(iii) 2x(3x-4) = 0: \boxed{(0, 6) \text{ and } \left(\frac{4}{3}, \frac{98}{27}\right)} \quad \left(\frac{98}{27} \approx 3.63\right)$$

78



Calculator

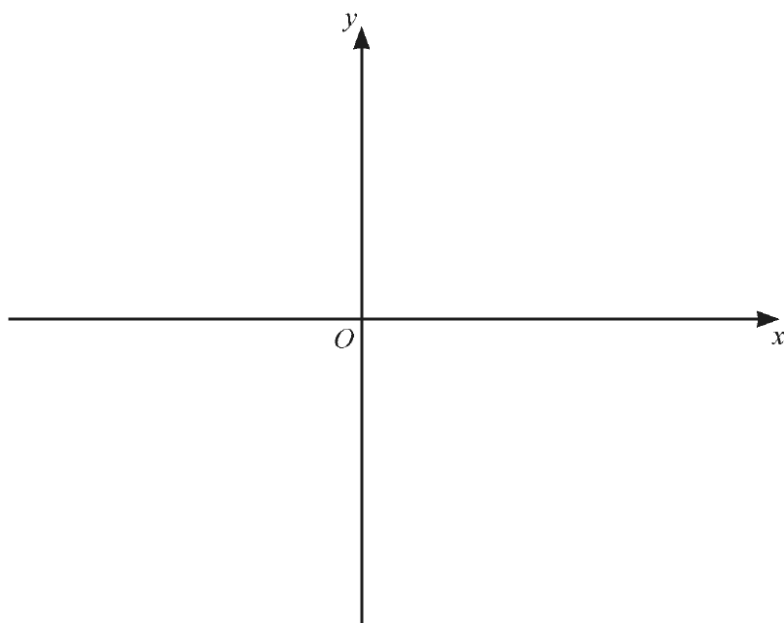
May/June 2021 · Paper 42 · Q9

- (a) (i) The equation $y = x^3 - 4x^2 + 4x$ can be written as $y = x(x-a)^2$.

Find the value of a .

$a = \dots\dots\dots$ [2]

- (ii) On the axes, sketch the graph of $y = x^3 - 4x^2 + 4x$, indicating the values where the graph meets the axes.



[4]

- (b) Find the equation of the tangent to the graph of $y = x^3 - 4x^2 + 4x$ at $x = 4$.
Give your answer in the form $y = mx + c$.

$y = \dots\dots\dots$ [7]

78 May/June 2021 · Paper 42 · Q9

(a)(i) $x(x^2 - 4x + 4) = x(x - 2)^2$: $a = 2$ (ii) Cubic through O , touching the x -axis at $x = 2$

(b) $\frac{dy}{dx} = 3x^2 - 8x + 4 = 20$ at $x = 4$; $y = 64 - 64 + 16 = 16$

$y - 16 = 20(x - 4) \Rightarrow y = 20x - 64$

79



Calculator

May/June 2021 · Paper 43 · Q10

$f(x) = 3x - 2$

$g(x) = 5x - 7$

$h(x) = x^2 + x$

$j(x) = 3^x$

(a) Find

(i) $f(2)$,

..... [1]

(ii) $g(2)$,

..... [1]

(iii) $gf(2)$.

..... [1]

(b) Find $f^{-1}(x)$.

$f^{-1}(x) = \dots\dots\dots [2]$

(c) Find $hf(x)$, giving your answer in the form $ax^2 + bx + c$.

..... [3]

(d) Find the derivative of $h(x)$.

..... [1]

(e) (i) Find x when $j^{-1}(x) = 4$.

$x = \dots\dots\dots [1]$

(ii) Simplify $j^{-1}j(x)$.

..... [1]

79 *May/June 2021 · Paper 43 · Q10*

(a)(i) $\boxed{4}$ (ii) $\boxed{3}$ (iii) $g(4) = \boxed{13}$ (b) $\boxed{\frac{x+2}{3}}$

(c) $(3x-2)^2 + (3x-2) = \boxed{9x^2 - 9x + 2}$ (d) $\boxed{2x+1}$ (e)(i) $x = 3^4 =$

$\boxed{81}$ (ii) $\boxed{x}$

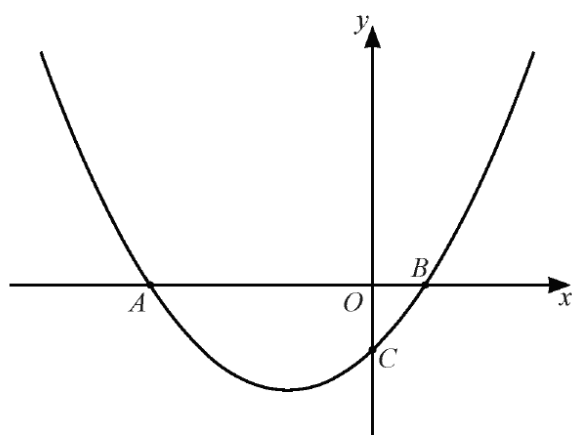
80

(a)



Calculator

Oct/Nov 2020 · Paper 41 · Q10

NOT TO
SCALE

The diagram shows a sketch of the curve $y = x^2 + 3x - 4$.

(i) Find the coordinates of the points A , B and C .

A (.....,)

B (.....,)

C (.....,) [4]

(ii) Differentiate $x^2 + 3x - 4$.

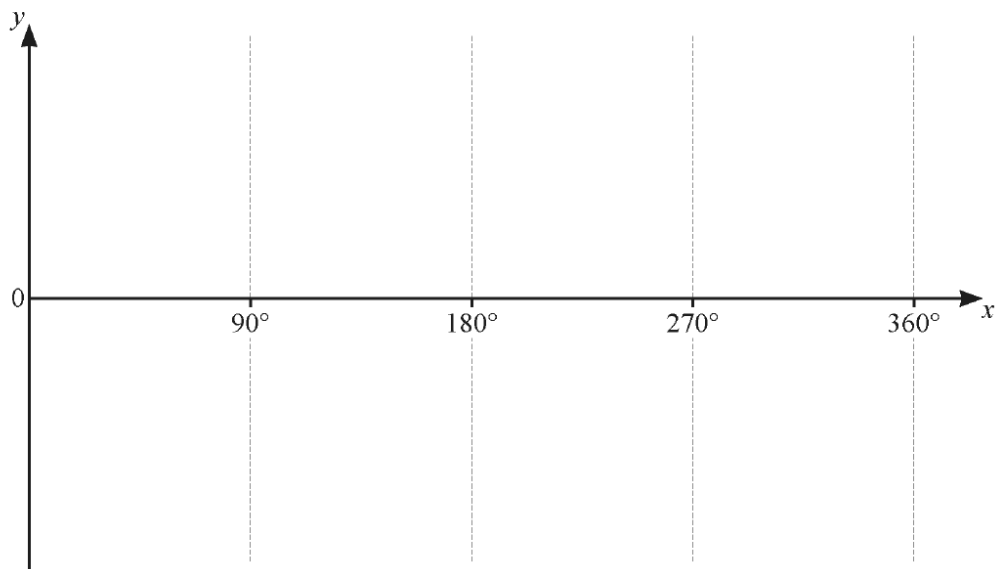
..... [2]

(iii) Find the equation of the tangent to the curve at the point $(2, 6)$.

..... [3]

(b)

Question continues on the next page →



(i) On the diagram, sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$. [2]

(ii) Solve the equation $5 \tan x = -7$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$ [3]

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(a)(i) $x^2 + 3x - 4 = (x + 4)(x - 1)$: $A(-4, 0), B(1, 0), C(0, -4)$ (ii)

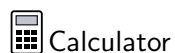
$2x + 3$

(iii) gradient at $x = 2$: 7; $y - 6 = 7(x - 2) \Rightarrow y = 7x - 8$

(b)(i) $\tan x$: through $0^\circ, 180^\circ, 360^\circ$, asymptotes at 90° and 270°

(ii) $\tan x = -1.4$: $x = 125.5^\circ$ or 305.5°

81



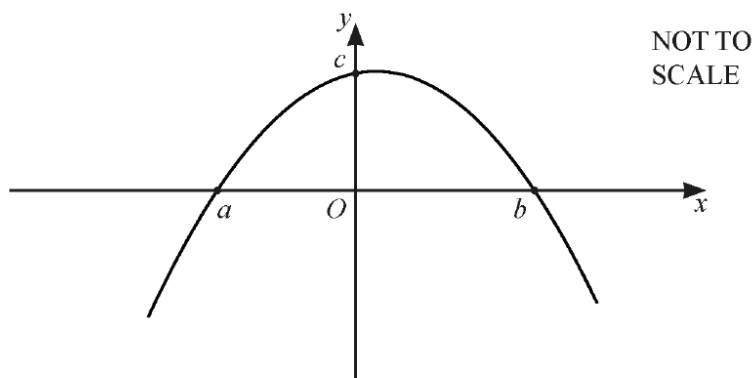
Calculator

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- (a) (i) Factorise $24 + 5x - x^2$.

..... [2]

- (ii) The diagram shows a sketch of $y = 24 + 5x - x^2$.



Work out the values of a , b and c .

$a =$

$b =$

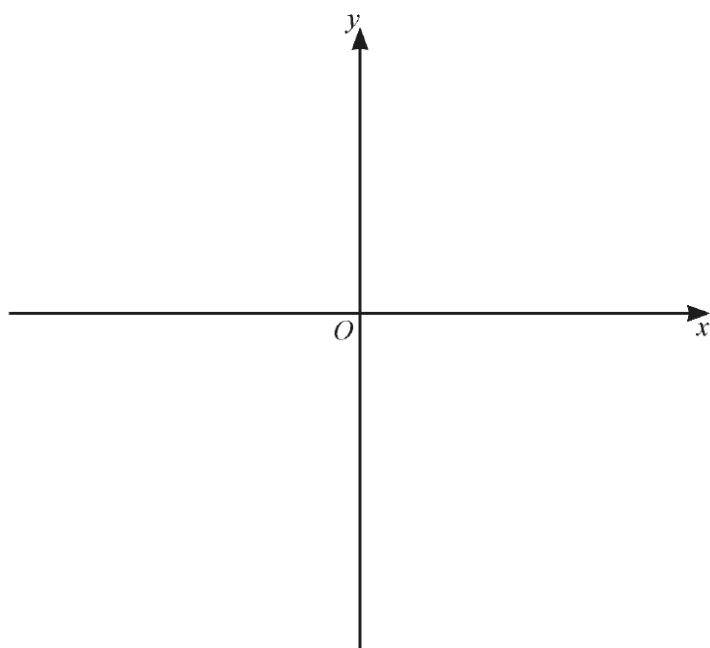
$c =$ [3]

- (iii) Calculate the gradient of $y = 24 + 5x - x^2$ at $x = -1.5$.

..... [3]

- (b) (i) On the diagram, sketch the graph of $y = (x+1)(x-3)^2$.
Label the values where the graph meets the x -axis and the y -axis.

Question continues on the next page →



[4]

(ii) Write $(x+1)(x-3)^2$ in the form $ax^3 + bx^2 + cx + d$.

..... [3]

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(a)(i) $(8-x)(3+x)$ (ii) $a = -3, b = 8, c = 24$ (iii) $\frac{dy}{dx} =$

$5 - 2x = 5 + 3 = 8$

(b)(i) Cubic crossing the x -axis at -1 , touching it at 3 (minimum),
y-intercept 9

(ii) $(x+1)(x^2 - 6x + 9) = x^3 - 5x^2 + 3x + 9$

82



Calculator

May/June 2020 · Paper 41 · Q10

(a) A rhombus $ABCD$ has a diagonal AC where A is the point $(-3, 10)$ and C is the point $(4, -4)$.

(i) Calculate the length AC .

..... [3]

(ii) Show that the equation of the line AC is $y = -2x + 4$.

[2]

(iii) Find the equation of the line BD .

..... [4]

(b) A curve has the equation $y = x^3 + 8x^2 + 5x$.

(i) Work out the coordinates of the two turning points.

(.....,) and (.....,) [6]

(ii) Determine whether each of the turning points is a maximum or a minimum.
Give reasons for your answers.

[3]

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$$(a)(i) AC = \sqrt{7^2 + 14^2} = \sqrt{245} = \boxed{15.7} \quad (ii) m = \frac{-4 - 10}{4 - (-3)} = -2;$$

$$10 = -2(-3) + c \Rightarrow c = 4 \checkmark$$

(iii) BD is perpendicular to AC through the midpoint $(0.5, 3)$: gradient

$$\frac{1}{2}: \boxed{y = \frac{1}{2}x + 2.75}$$

$$(b)(i) \frac{dy}{dx} = 3x^2 + 16x + 5 = (3x + 1)(x + 5) = 0: \boxed{(-5, 50)} \text{ and}$$

$$\boxed{\left(-\frac{1}{3}, -\frac{22}{27}\right)}$$

$$(ii) \frac{d^2y}{dx^2} = 6x + 16: \text{ at } x = -5: -14 < 0 \text{ **maximum**; at } x = -\frac{1}{3}: 14 > 0$$

minimum

83

Calculator

May/June 2020 · Paper 42 · Q10

(a) $y = x^4 - 4x^3$

- (i) Find the value of
- y
- when
- $x = -1$
- .

$$y = \dots\dots\dots [2]$$

- (ii) Find the two stationary points on the graph of
- $y = x^4 - 4x^3$
- .

$$(\dots\dots\dots, \dots\dots\dots)$$

$$(\dots\dots\dots, \dots\dots\dots) [6]$$

(b) $y = x^p + 2x^q$

$$\frac{dy}{dx} = 11x^{10} + 10x^4, \text{ where } \frac{dy}{dx} \text{ is the derived function.}$$

Find the value of p and the value of q .

$$p = \dots\dots\dots$$

$$q = \dots\dots\dots [2]$$

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(a)(i) $y = 1 + 4 = \boxed{5}$ (ii) $4x^3 - 12x^2 = 4x^2(x - 3) = 0$: $\boxed{(0, 0) \text{ and } (3, -27)}$

(b) $px^{p-1} = 11x^{10} \Rightarrow \boxed{p = 11}$; $2qx^{q-1} = 10x^4 \Rightarrow \boxed{q = 5}$